

```
In [2]: #Strings are objects of the str class.
S1=" " #Creates an Empty string Object
S2="Hello" #Creates a String Object for Hello
print(S1)
print(S2)
```

Hello

```
In [1]: S1=str() #Creates an Empty string Object
S2=str("Hello") #Creates a String Object for Hello
print(S1)
print(S2)
```

Hello

```
In [3]: a = "PYTHON"
print(len(a)) #Return Length i.e. number of characters in string a

print(min(a)) #Return smallest character present in a string

print(max(a))
```

6
H
Y

```
In [5]: #Accessing Characters
#S[-3]=S[Len(S)-3]=S[10-3]=S[7]
a = "P Y T H O N"
print(len(a))
print(a[-4])
print(a[-6])
print(a[-8])
```

9
H
T

```
In [4]: #TRAVERSING STRING WITH for AND while LOOP
S="India"
for ch in S:
    print(ch,end=" ")
```

I n d i a

```
In [7]: S="India"
for ch in range(0,len(S),1):#Traverse each Second character
    print(S[ch],end=" ")
```

I n d i a

```
In [10]: S="India"
index=0
while index<len(S):
    print(S[index],end=" ")
    index=index+1
```

I n d i a

```
In [11]: #strings are immutable sequences of characters
Str1="I Love Python"
Str1[0]="U"
print(Str1)
```

```
-----
TypeError                                Traceback (most recent call last)
<ipython-input-11-2c05febea8b1> in <module>
      1 #strings are immutable sequences of characters
      2 Str1="I Love Python"
----> 3 Str1[0]="U"
      4 print(Str1)
```

TypeError: 'str' object does not support item assignment

```
In [12]: #If you want to change the existing string, the best way is to create a new string that is a
#variation of the original string.
Str1="I Love Python"
print(Str1[1:])
Str2="U"+Str1[1:]
print(Str2)
```

Love Python
U Love Python

```
In [16]: Str1="I Love Python"
print(Str1)
#Str2=Str[0:1]
Str2 = Str1[0:2] + "l" + Str1[3:7] + "p" + Str1[8:]
print(Str2)
print(Str1[:2]+"l"+ Str1[3:7] + "p"+Str2[8:])

#I Love python
```

I Love Python

```
I love python
I love python
```

```
In [18]: str1="Hello"
         #str2="Hello"
         str2=str1
         print(id(str1))
         print(id(str2))
```

```
2347271368880
2347271368880
```

```
In [14]: #String Slicing Operator [start: end]
         a2="Hello hi"
         print(a2)
         print(a2[1:4])
         print(a2[-3:0:-1])
         print(a2[-1:-4:-1])
         print(a2[-1:-4:1])
         print(a2[-3:-1])
         print(a2[-3:])
```

```
Hello hi
ell
olle
ih
```

```
h
hi
```

```
In [15]: b1="i love my india"
         print(b1)
         print(b1[0:2])
         print(b1[:2])
         print(b1[::2])
         print(b1[::-2])
         print(b1[-1:-4:-2])
```

```
i love my india
i
i
ilv yida
adiy vli
ad
```

```
In [7]: #String Operator
         t1="Hello"
         t2="Hi"
         t3="Bye"
         #print(t1-t2)
         print(t1+t2) # concatenation of both the String+
         print(t1+t3)
         print(4*t2) # Replication of the String by *
         print(2*t3)
         print ("l l" in t1)
         t5='hi'
         print (t2 is t5)
```

```
HelloHi
HelloBye
HiHiHiHi
ByeBye
False
False
```

```
In [23]: #String Comparison
         #Operators such as ==,<,>,<=,>=and != are used to compare the strings
         S1="abcd"
         S2="ABCD"
         S3="AbCd"
         S4="A C "
         print(S1>S2)
         print(S1>S3)
         print(S2>S3)
         print(S2>S4)
```

```
True
True
False
True
```

```
In [8]: s="10"
         print(s.upper())
         print(s.lower())
         print(s.isdigit())
```

```
10
10
True
```

```
In [ ]:
```

```
In [2]: s="21Msd7050"
         print(s)
         d1=0
         c1=0
```

```
s1=0
for i in s:
    if ord(i)>=48 and ord(i)<=57:
        d1=d1+1
    elif ord(i)>=65 and ord(i)<=90:
        c1=c1+1
    elif ord(i)>=97 and ord(i)<=122:
        s1=s1+1
print("number of digits:",d1)
print("number of capital letters:",c1)
print("number of small letters:",s1)
```

```
21Msd7050
number of digits: 6
number of capital letters: 1
number of small letters: 2
```

In []: